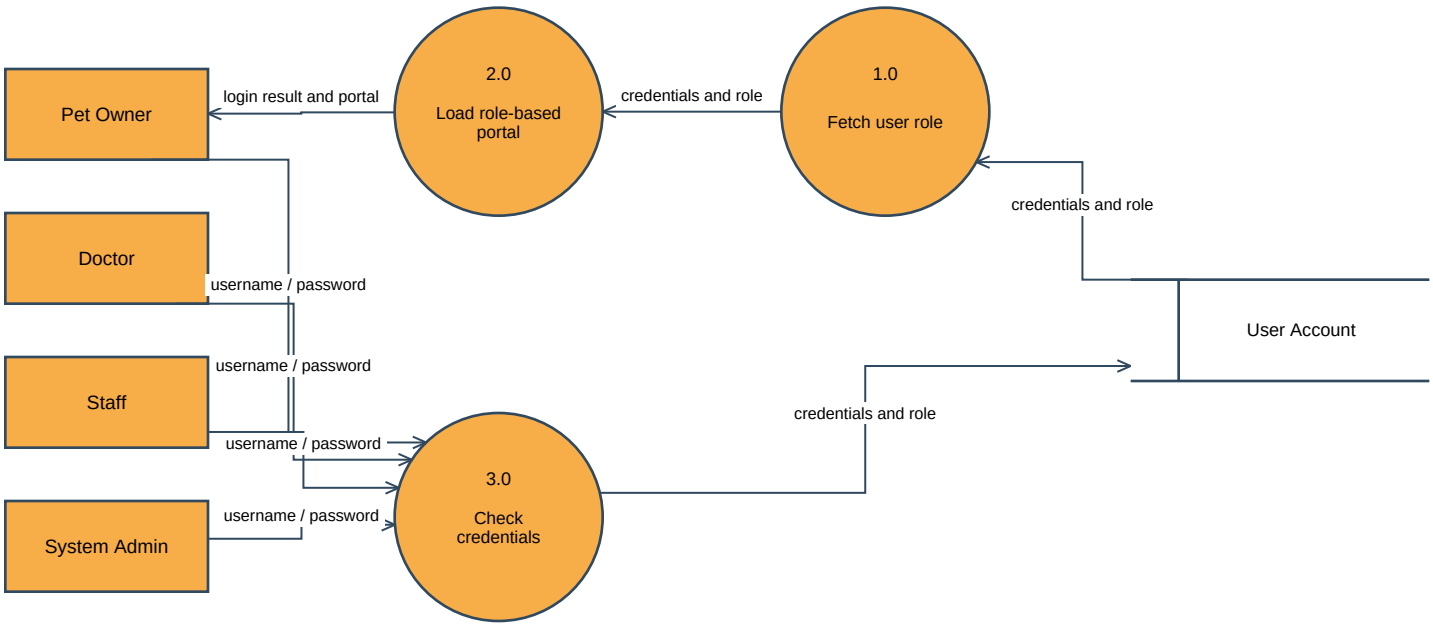
2. DFD Diagrams

Nway's Love Veterinary Clinic Application

The following diagrams use the same feature-by-feature structure as the reference report. Orange rectangles represent external users, orange circles represent numbered processes, open rectangles represent data stores, and labeled arrows represent data moving between them.

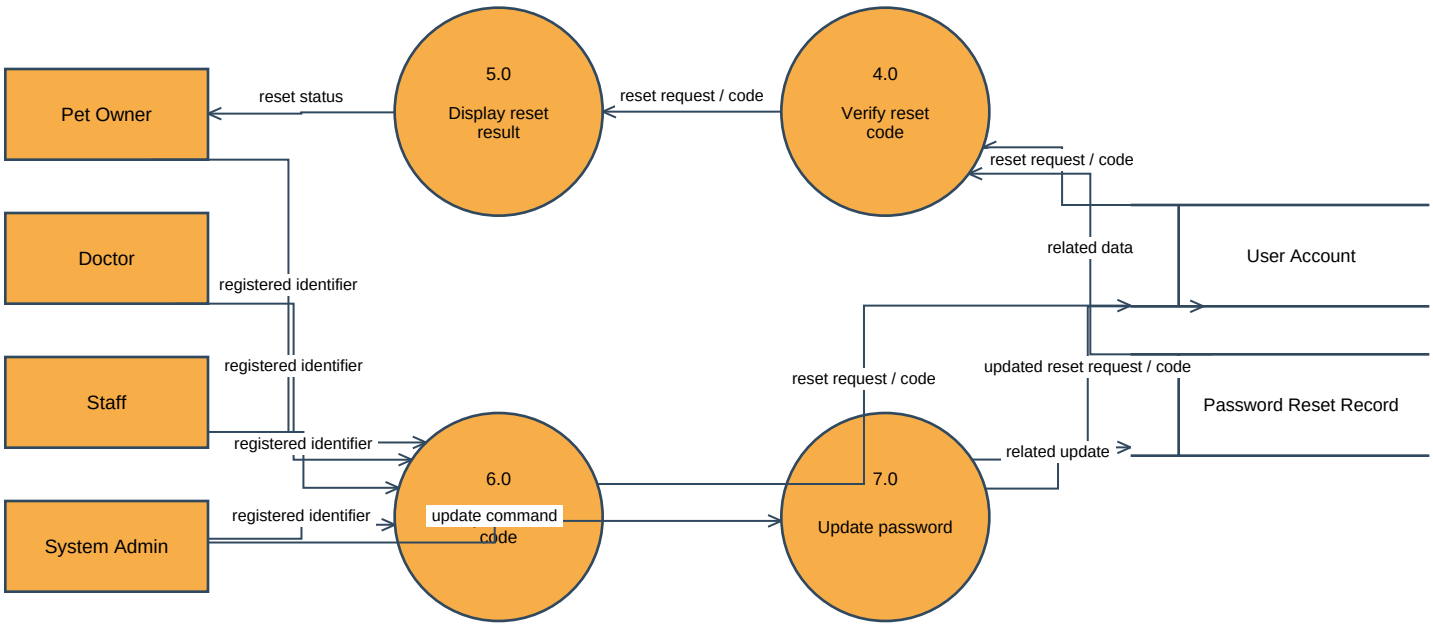
User Login and Role-Based Access

A user enters a username, email, or phone number and password. The application validates the credentials against the user account data, identifies the account role and status, and opens the correct Pet Owner, Doctor, Staff, or System Admin portal.



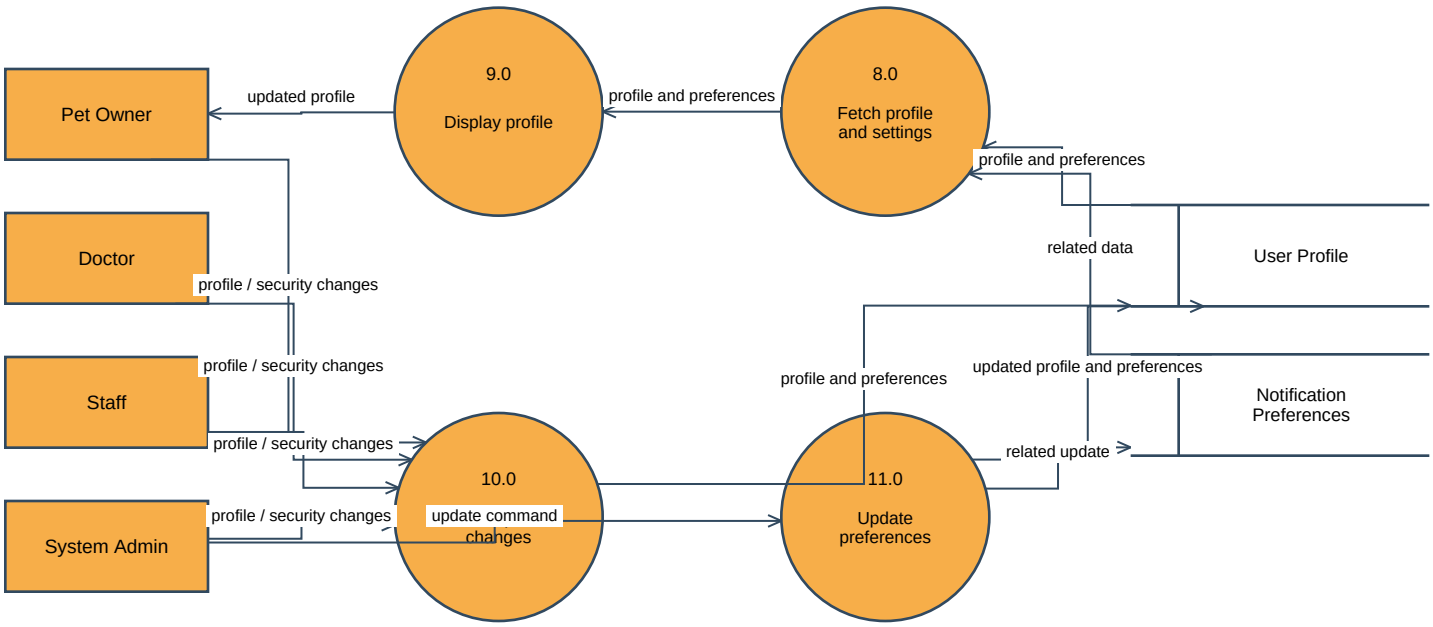
Password Recovery

A user requests password recovery using a registered identifier. The system locates the account, creates and sends a reset code, verifies the submitted code, and stores the new password after successful validation.



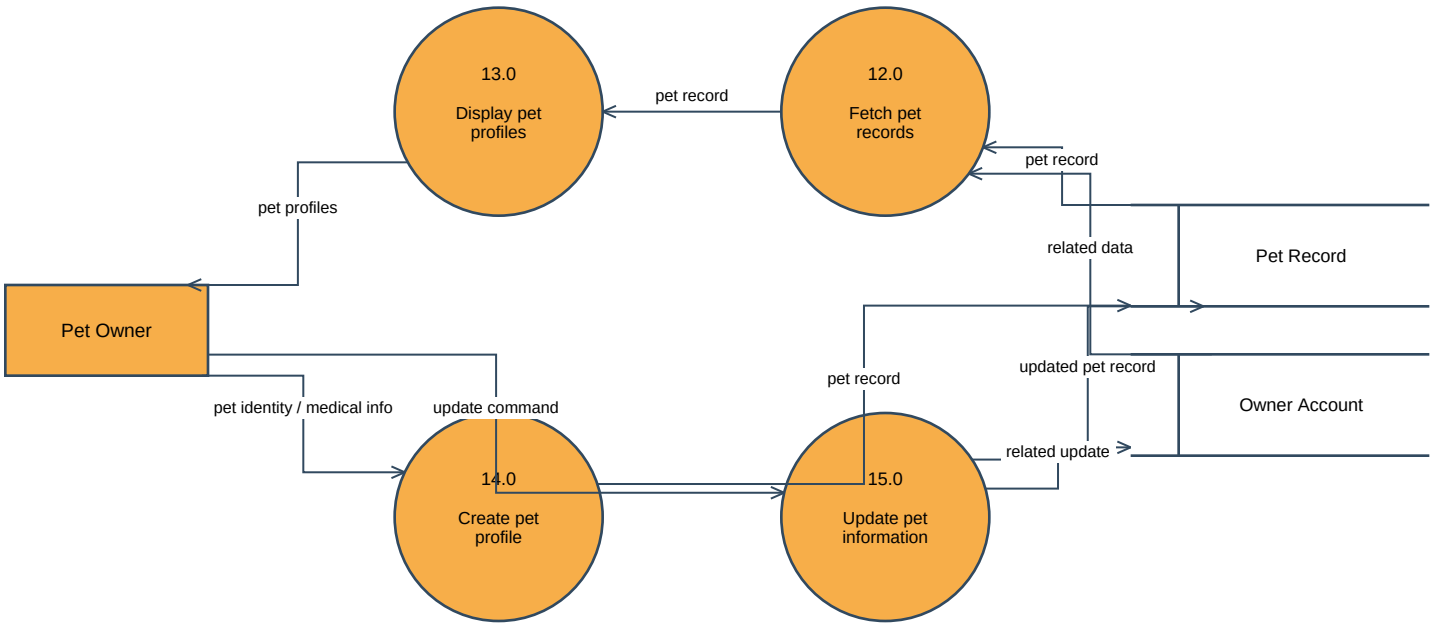
Profile, Security, and Notification Settings

Authenticated users view and update their profile, photo, contact information, password, security options, and role-specific notification preferences. Validated changes are saved and returned to the account interface.



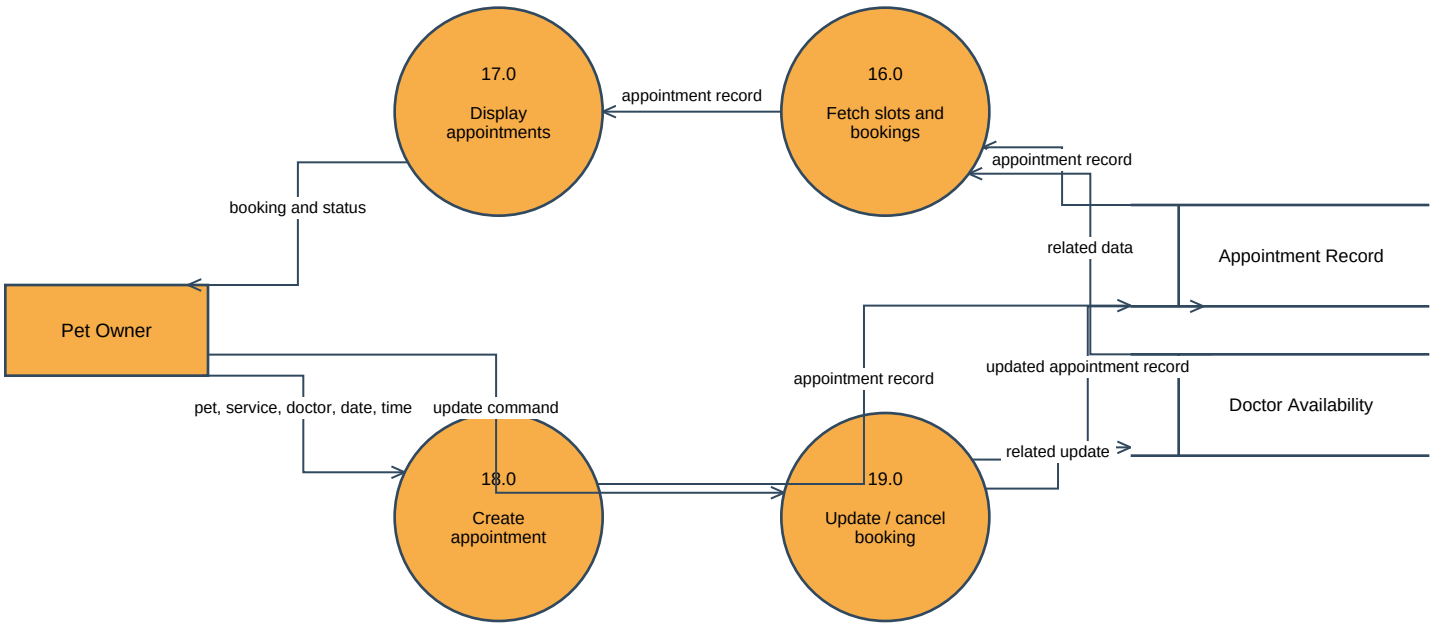
Pet Profile Management (Pet Owner)

The pet owner creates, views, and updates pet profiles containing the pet's name, species, breed, sex, weight, photo, vaccine information, and other medical details. Every pet record remains linked to the authenticated owner account.



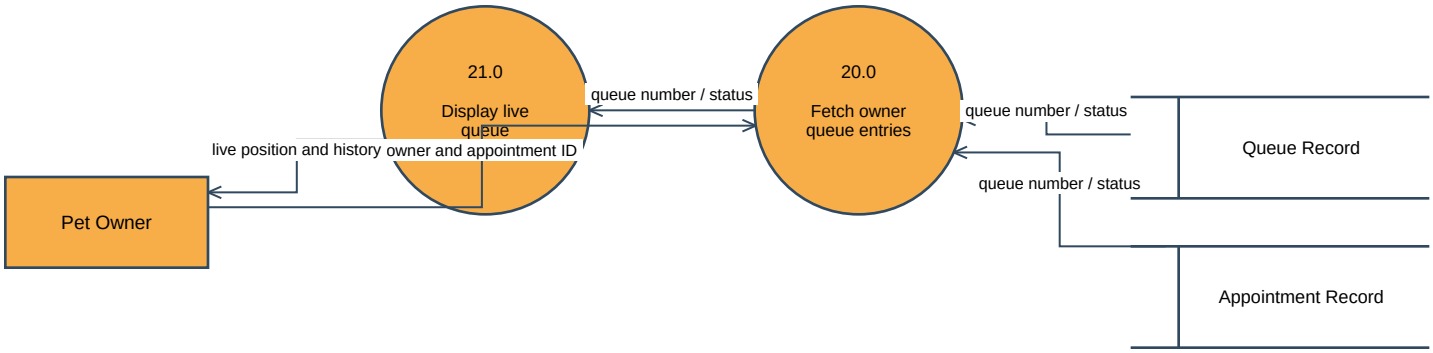
Appointment Booking and Tracking (Pet Owner)

A pet owner selects a pet, service, veterinarian, date, time, and reason. The application checks doctor availability, stores the booking, and displays appointment details and subsequent confirmation, rescheduling, cancellation, or completion status.



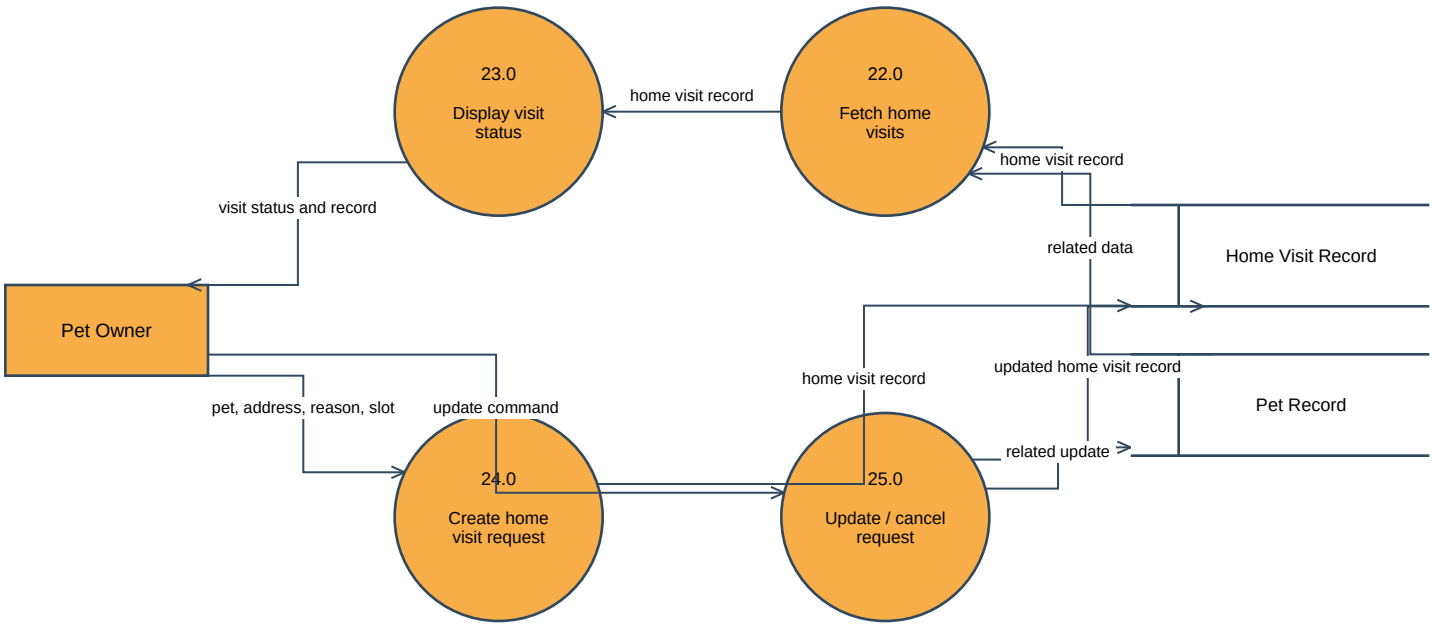
Queue Status and Queue History (Pet Owner)

The pet owner opens the queue screen to retrieve active and historical queue entries linked to the owner's appointments. The application displays the queue number, service group, doctor, current status, and previous queue activity.



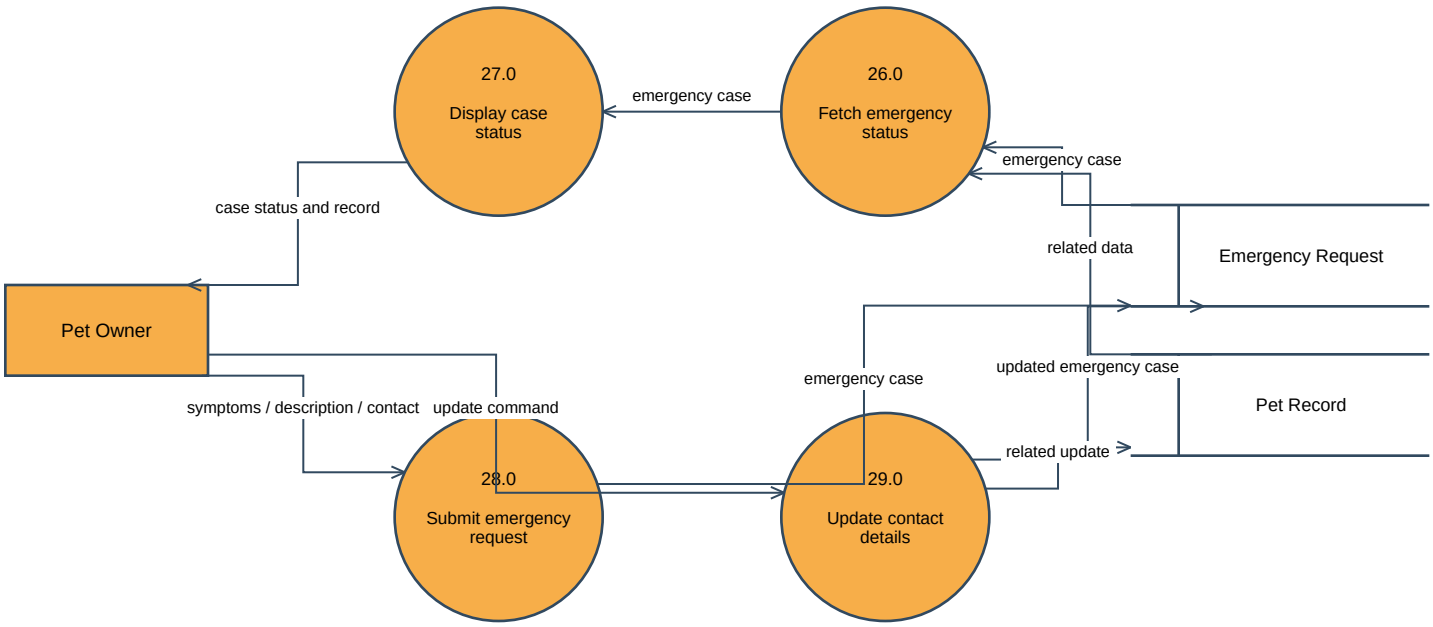
Home Visit Booking and Tracking (Pet Owner)

The pet owner submits a home-visit request with the selected pet, address, contact details, reason, preferred date, and time. The system stores the request and displays assignment, progress, completion, and the resulting home-visit medical record.



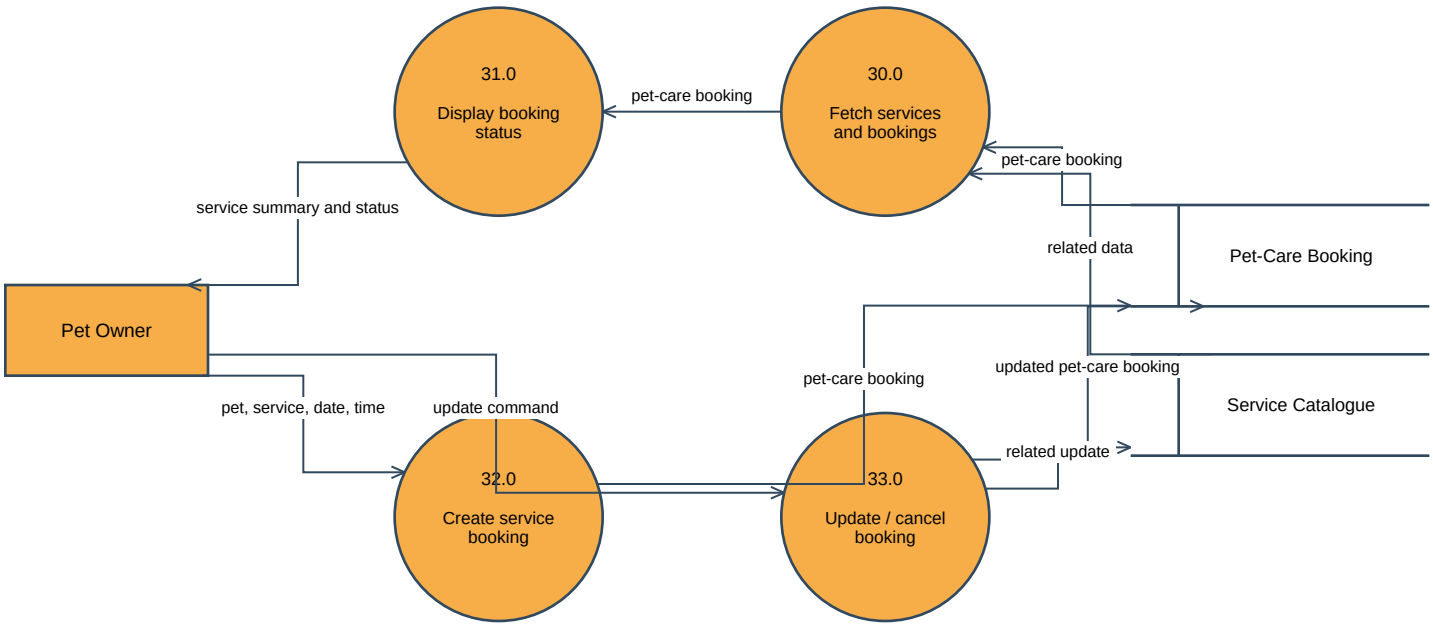
Emergency Service Request and Tracking (Pet Owner)

A pet owner reports an emergency by selecting a pet and providing symptoms, a description, and contact details. The application stores the case and displays staff review, acceptance, arrival, treatment, and the completed emergency medical record.



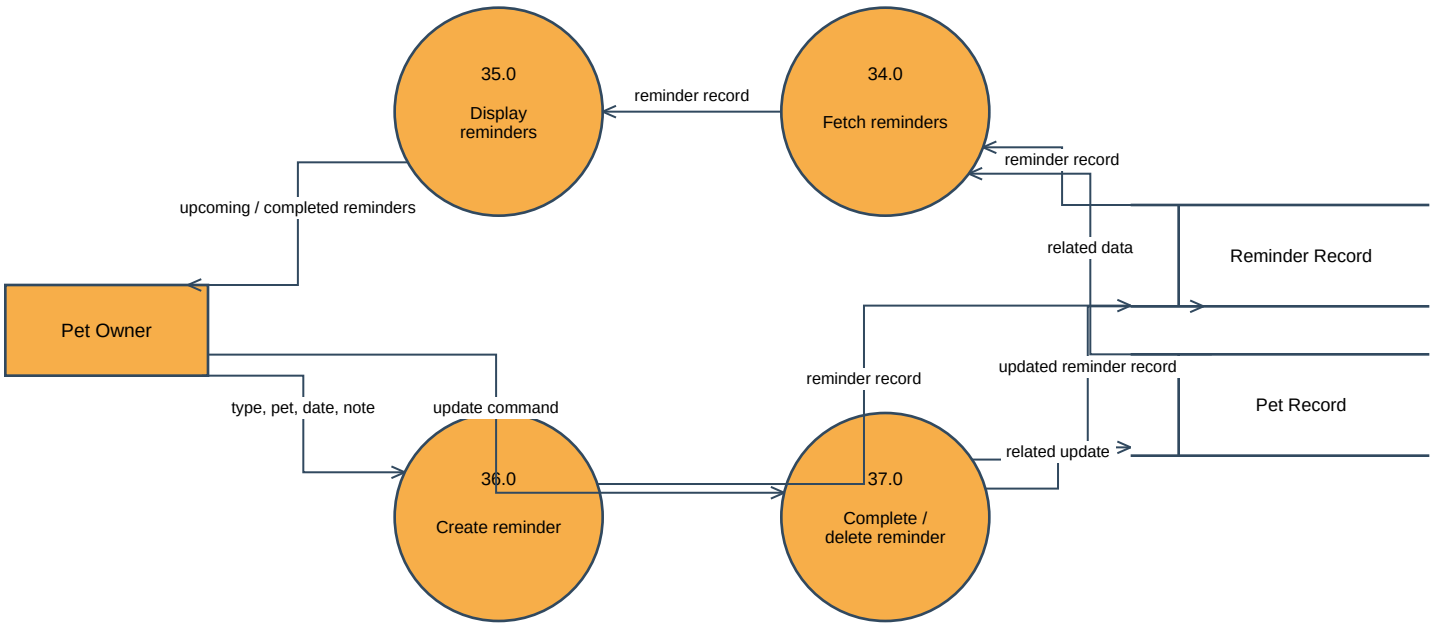
Pet-Care Service Booking (Pet Owner)

The pet owner selects a grooming, bathing, nail-trimming, boarding, or wellness service and submits the pet, date, and time information. The system stores the service booking and returns its summary and current status.



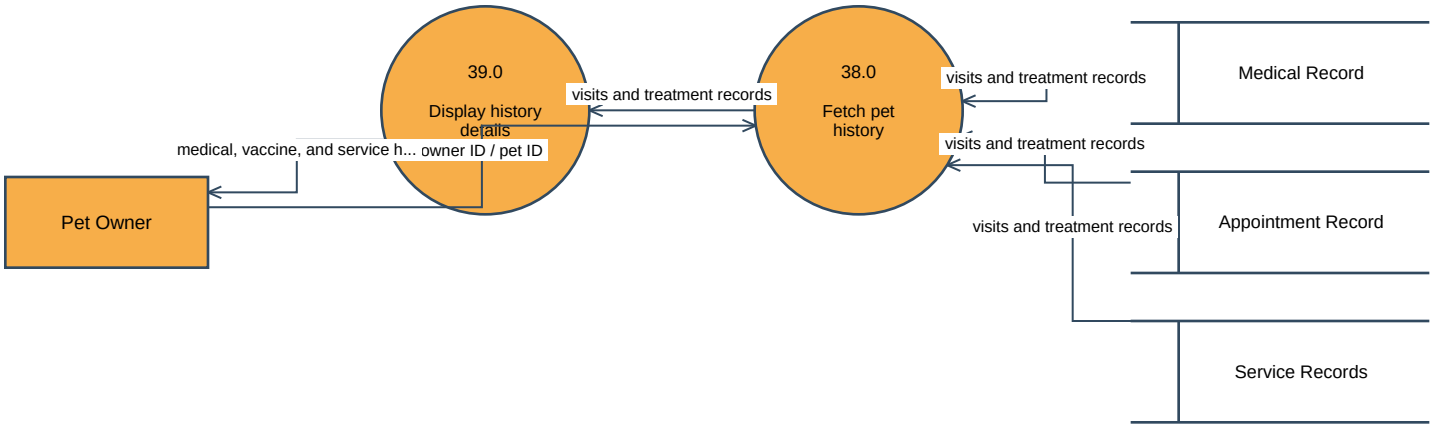
Pet Reminder Management (Pet Owner)

The pet owner creates reminders for medicine, vaccines, checkups, or other care tasks. The application stores each reminder, retrieves upcoming and completed items, and records completion or deletion commands.



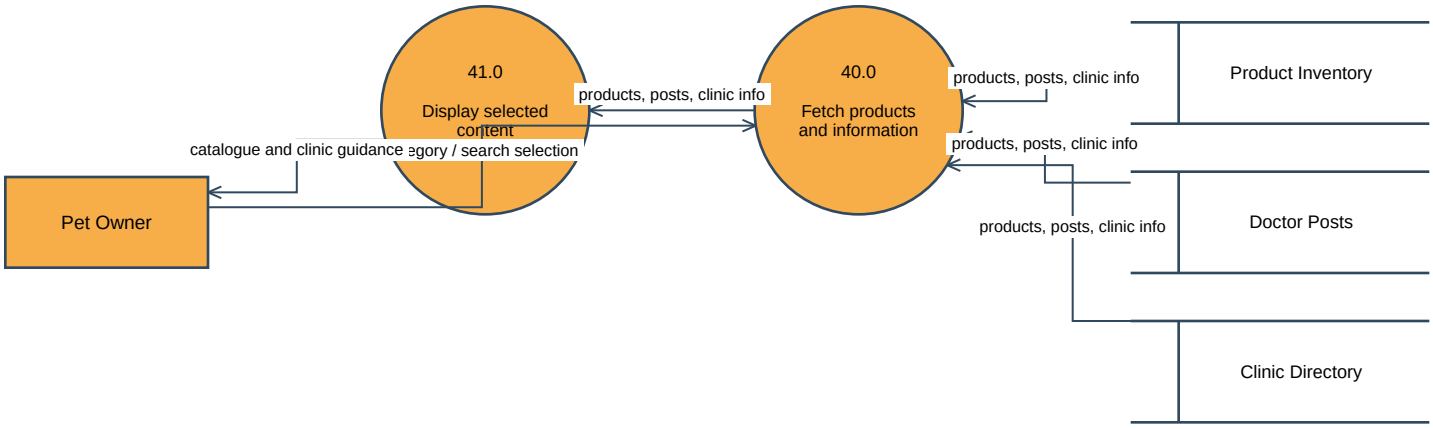
Medical and Treatment History Viewing (Pet Owner)

The pet owner requests a selected pet's history. The application combines clinic appointments, diagnoses, prescriptions, vaccinations, home visits, emergencies, pet-care services, and queue records and displays detailed historical information.



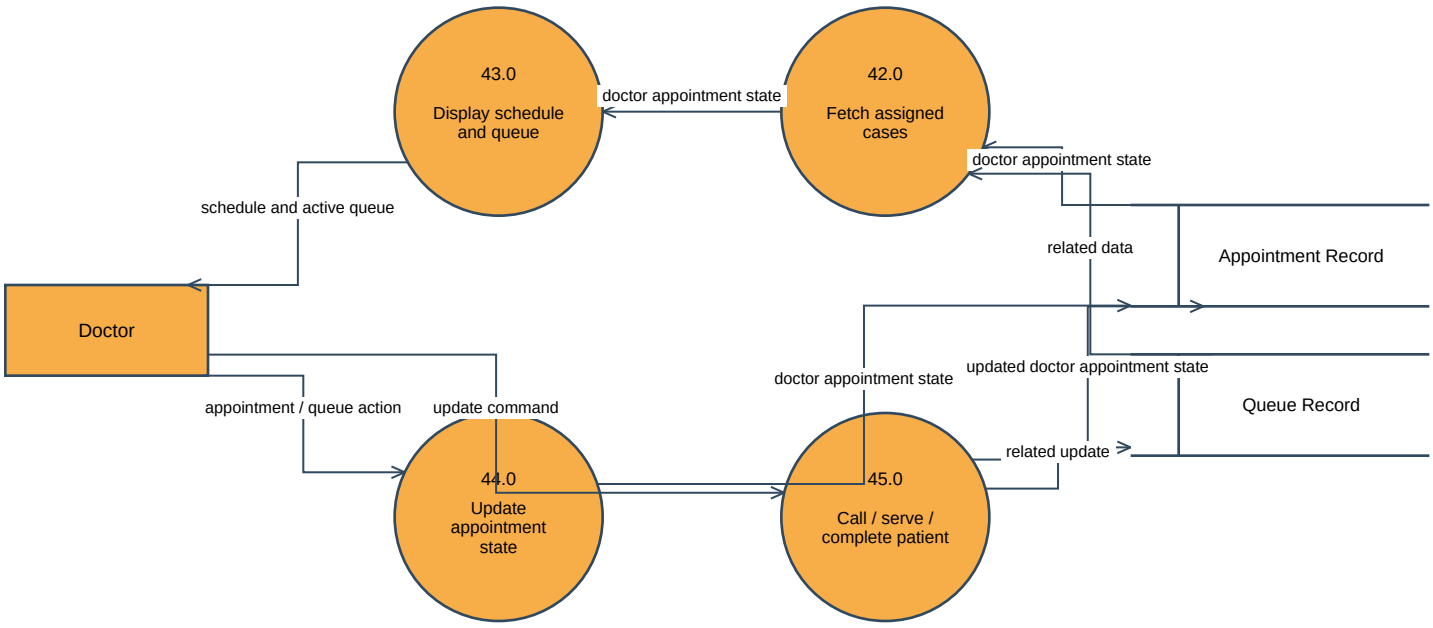
Products and Clinic Information Viewing (Pet Owner)

The pet owner browses products, doctor profiles, medical services, health posts, first-aid guides, clinic contact information, and available service categories. The application retrieves and displays the selected information without changing operational records.



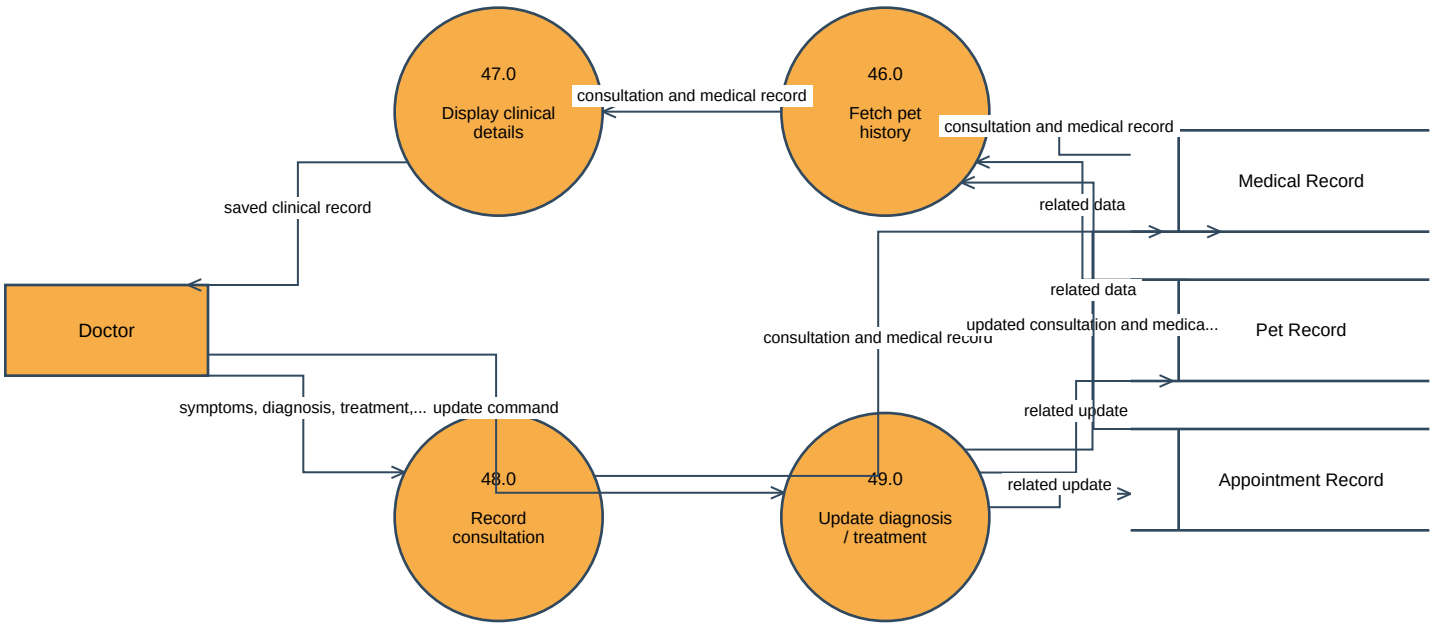
Doctor Appointment and Queue Management

The doctor retrieves assigned appointments and the live patient queue, opens patient details, and records workflow actions such as calling the next patient, starting a consultation, marking a patient missed, or completing the visit.



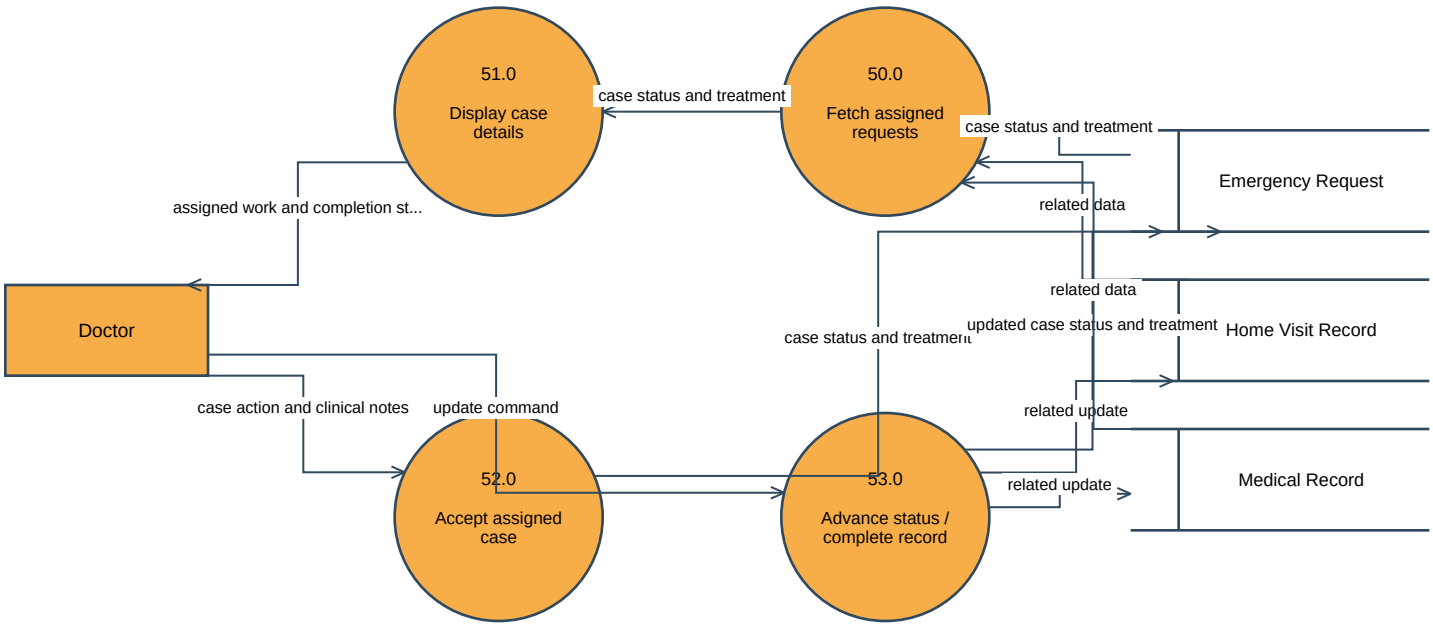
Doctor Consultation and Medical Record Update

During a consultation, the doctor reviews the pet profile and previous history, records symptoms and findings, enters diagnosis, treatment, prescription, vaccination, and follow-up details, and stores the completed medical record.



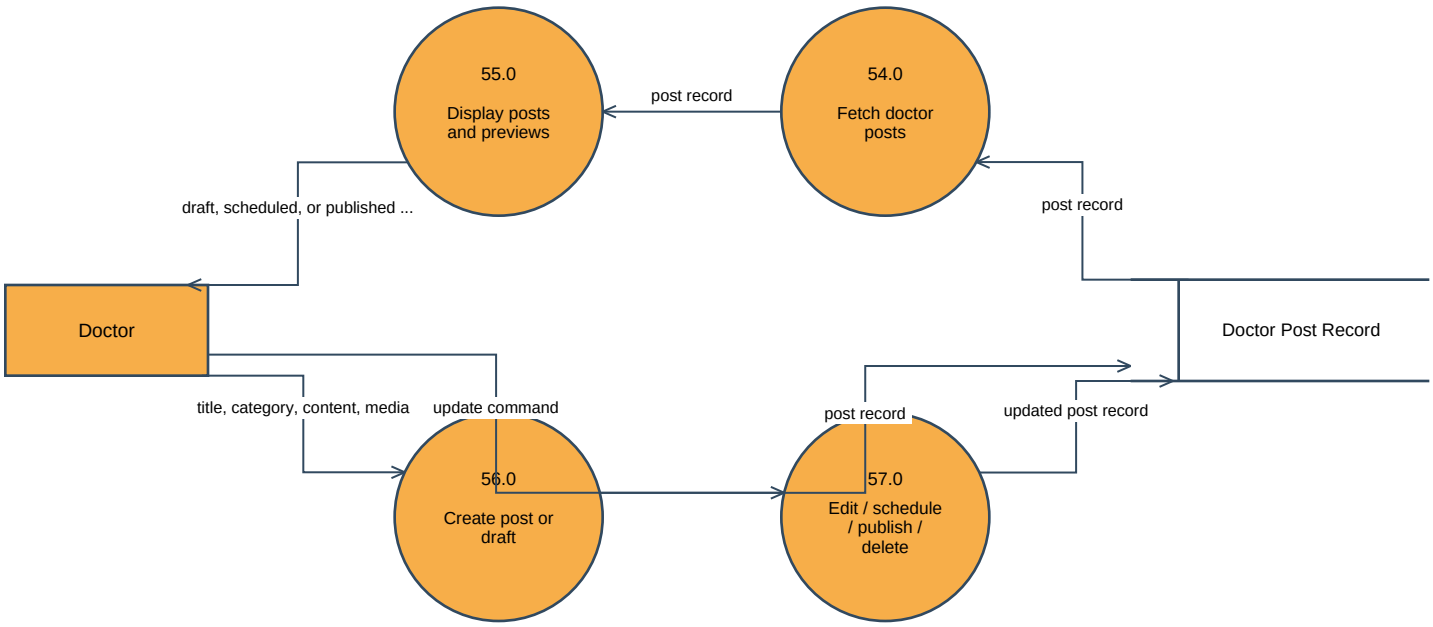
Doctor Emergency and Home-Visit Handling

The doctor views assigned emergency cases and home visits, reviews owner and pet details, advances the service status, contacts the owner when needed, and records treatment and completion information for the final medical record.



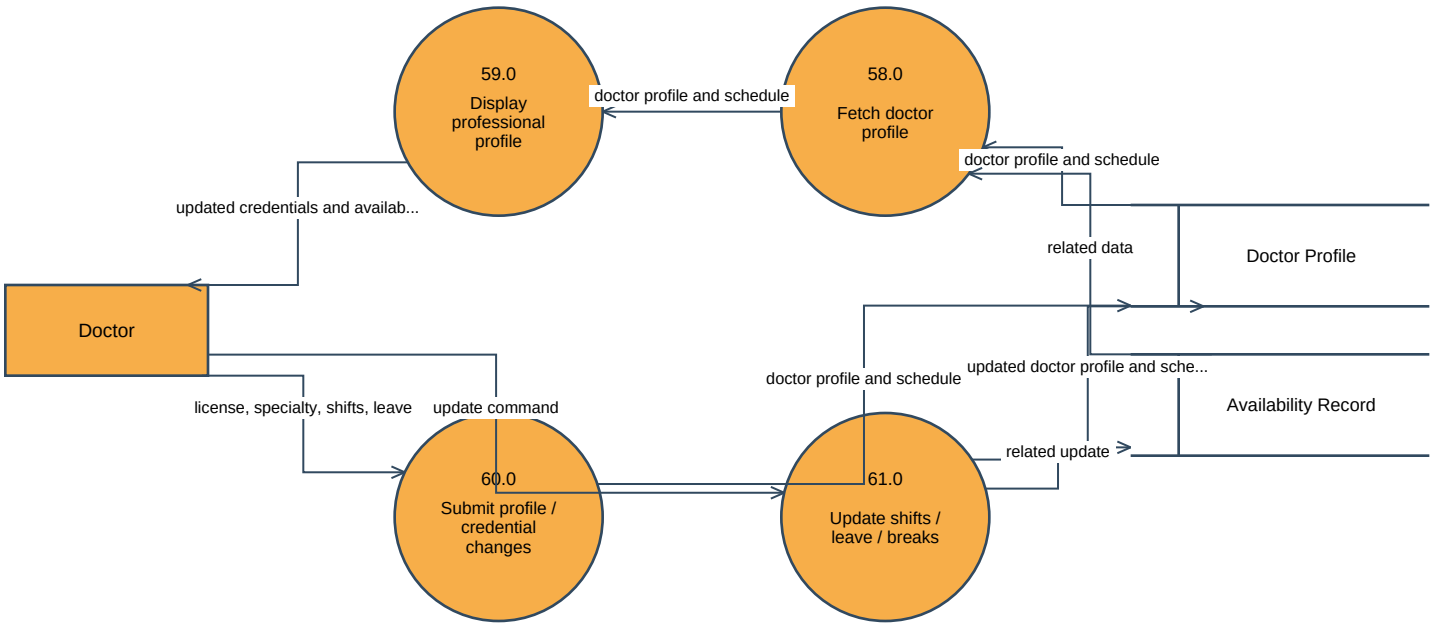
Doctor Health-Post Management

The doctor creates health-information posts containing a title, category, content, cover image, and attachments. Drafts can be previewed, edited, scheduled, published, or deleted, and published posts become available to pet owners and staff.



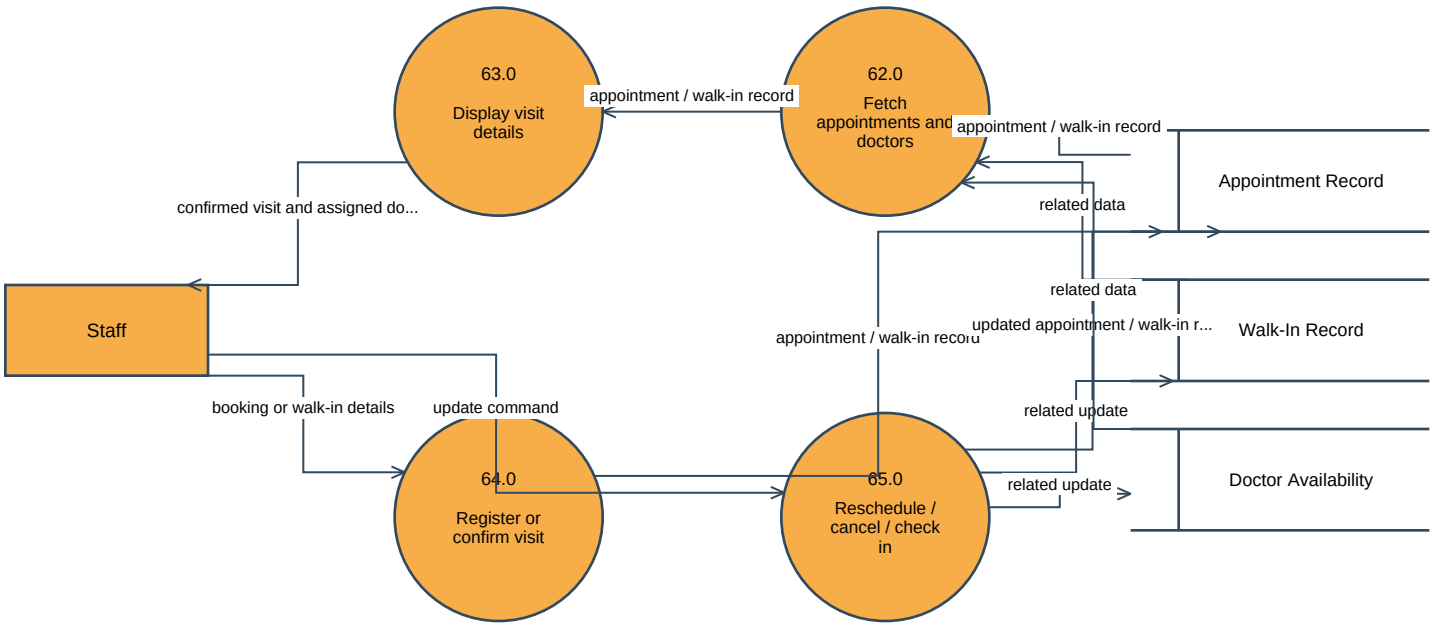
Doctor Profile, Credentials, and Availability

The doctor maintains contact details, specialty, qualifications, license information, weekly availability, shifts, breaks, and leave. The application validates and stores these details for appointment scheduling and public doctor profiles.



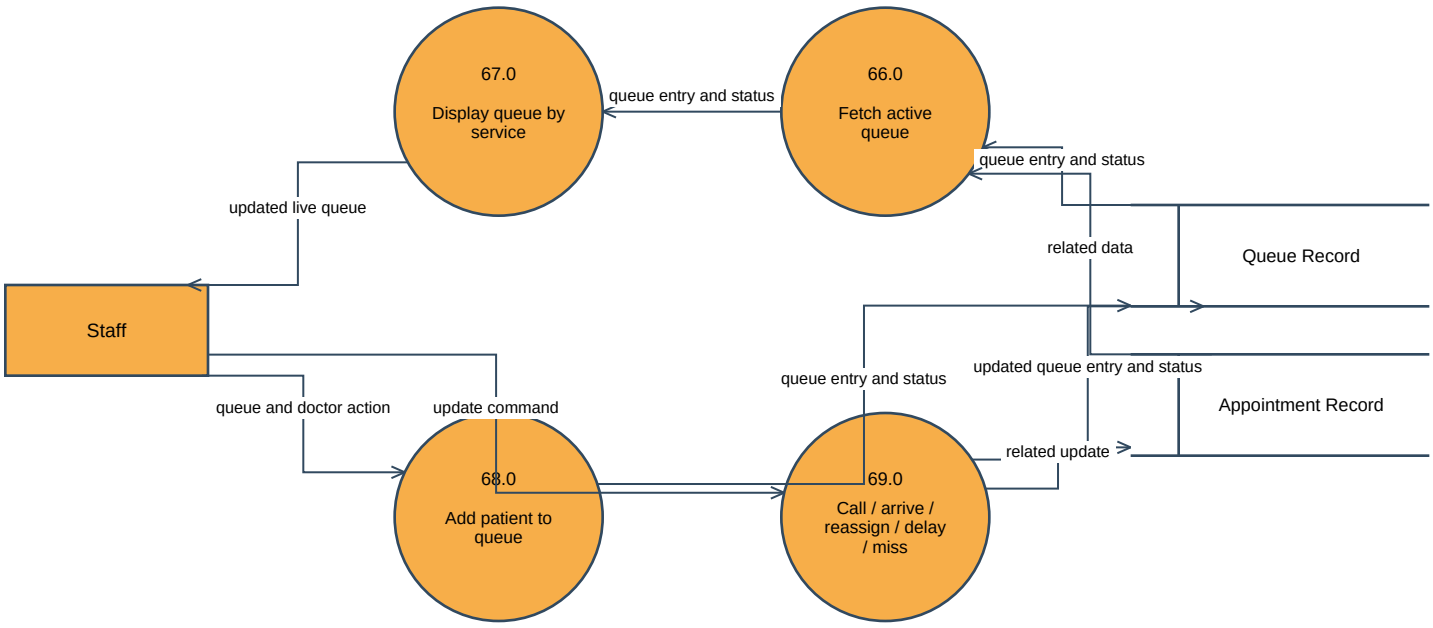
Staff Appointment and Walk-In Management

Staff retrieve appointment requests, confirm or reschedule bookings, assign doctors, contact owners, cancel visits, check in scheduled patients, and register walk-in patients before sending them to the live queue.



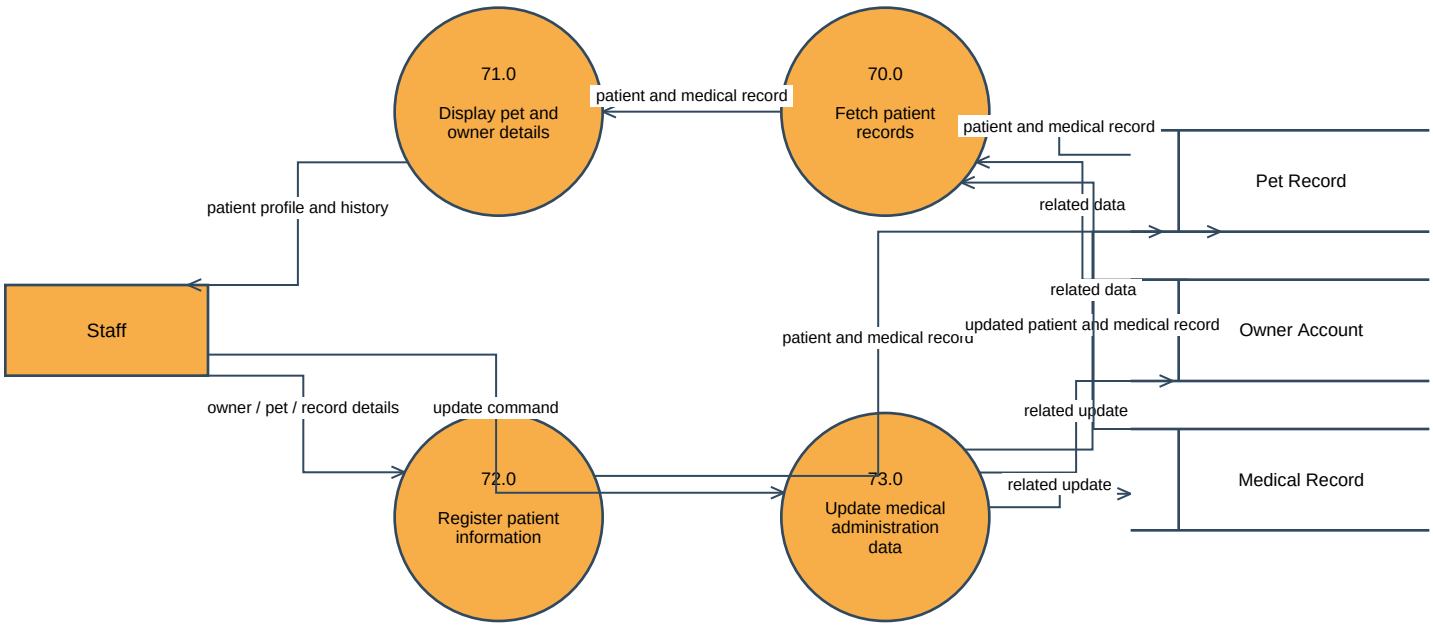
Staff Live Queue Management

Staff manage the real-time patient queue by service group and status. They add checked-in patients, call patients, mark arrival, reassign doctors, announce delays, mark missed visits, and monitor completion.



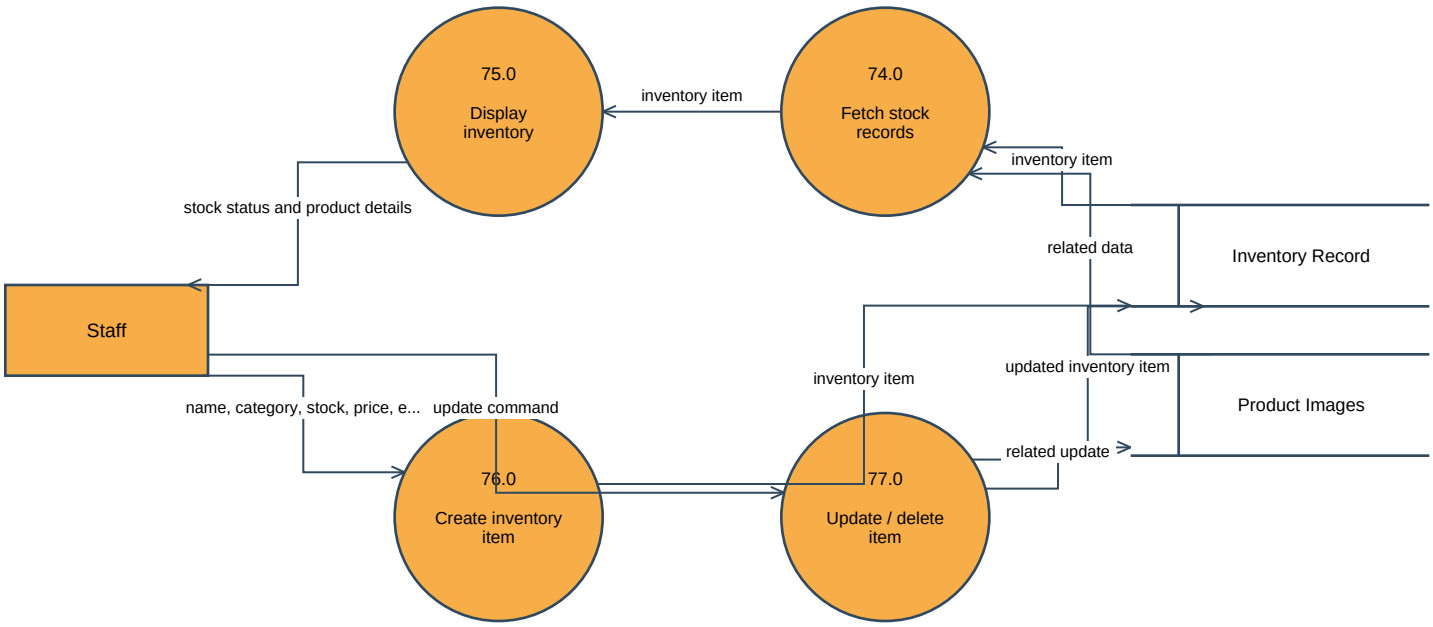
Staff Patient and Medical Record Management

Staff search pet owners and patients, open pet profiles, review treatment history, and maintain administrative medical-record details needed for clinic operations while preserving the doctor's clinical entries.



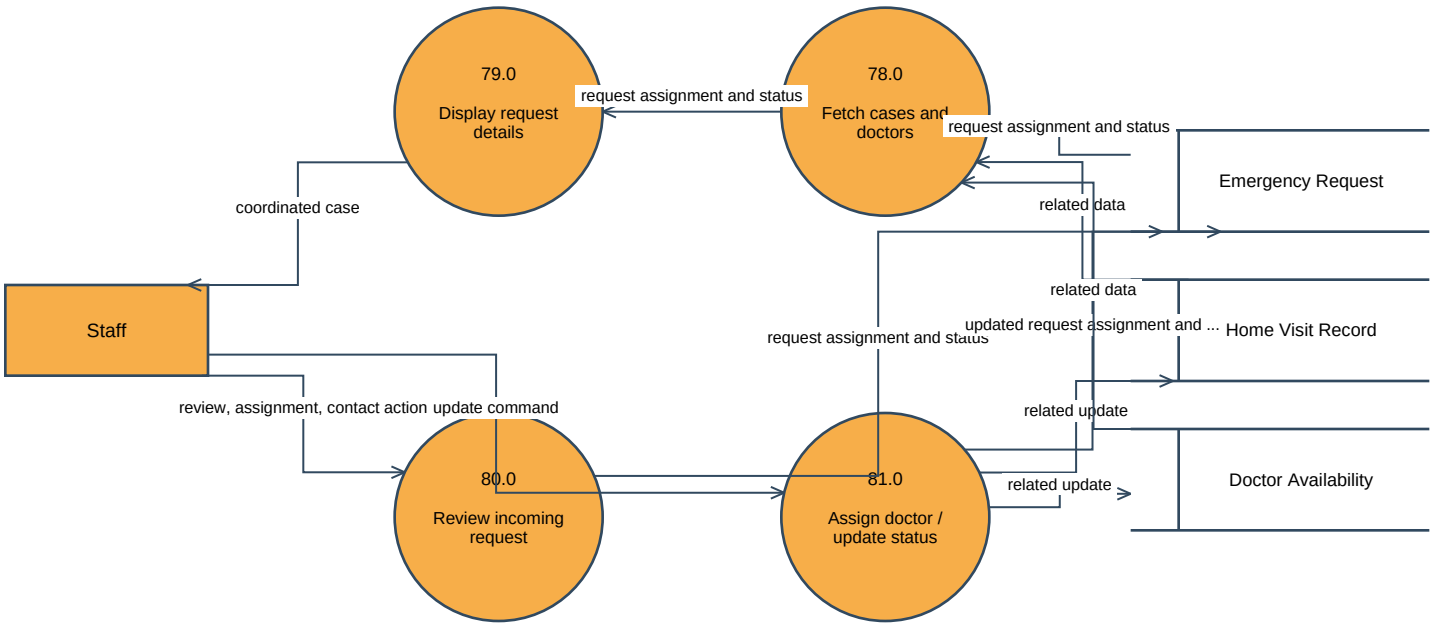
Staff Inventory Management

Staff create, view, search, update, and remove inventory items. Each record includes product images, category, brand, pet type, price, stock, reorder level, expiry date, and description, allowing the system to identify low-stock and expired items.



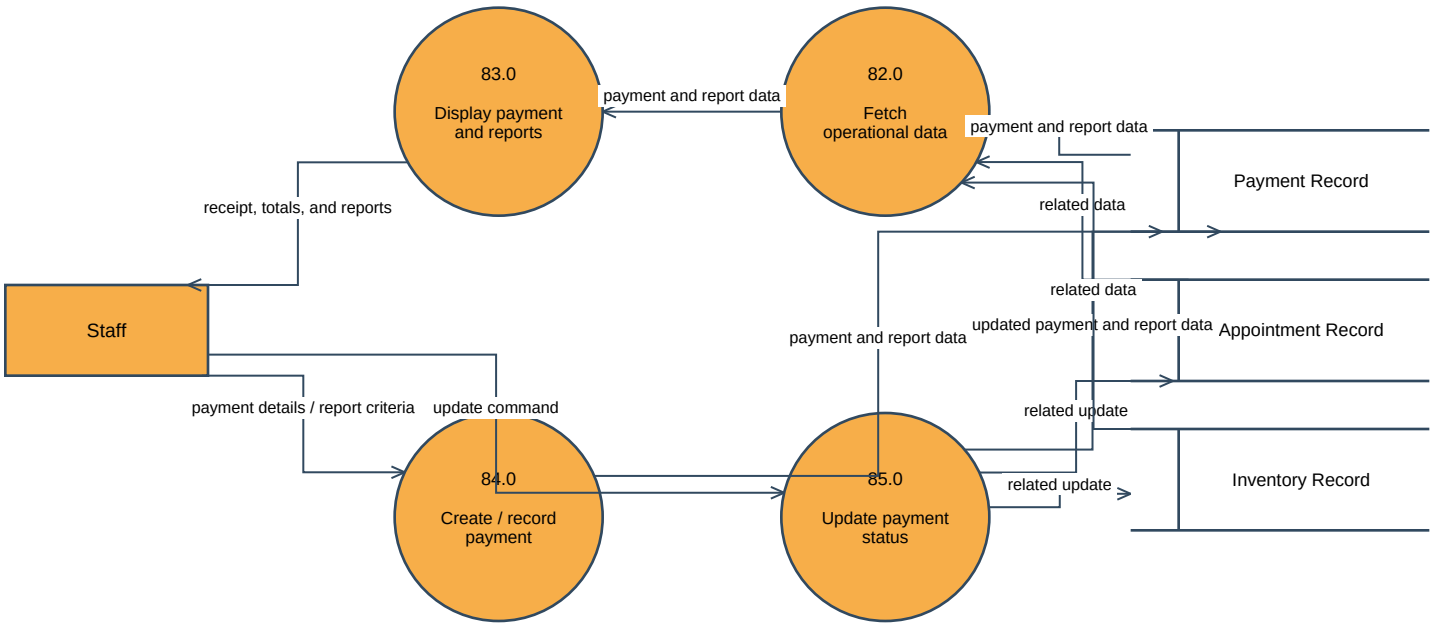
Staff Emergency and Home-Visit Coordination

Staff review incoming emergency and home-visit requests, contact the owner, select an available doctor, assign the case, and update acceptance, arrival, scheduling, and completion-related administrative status.



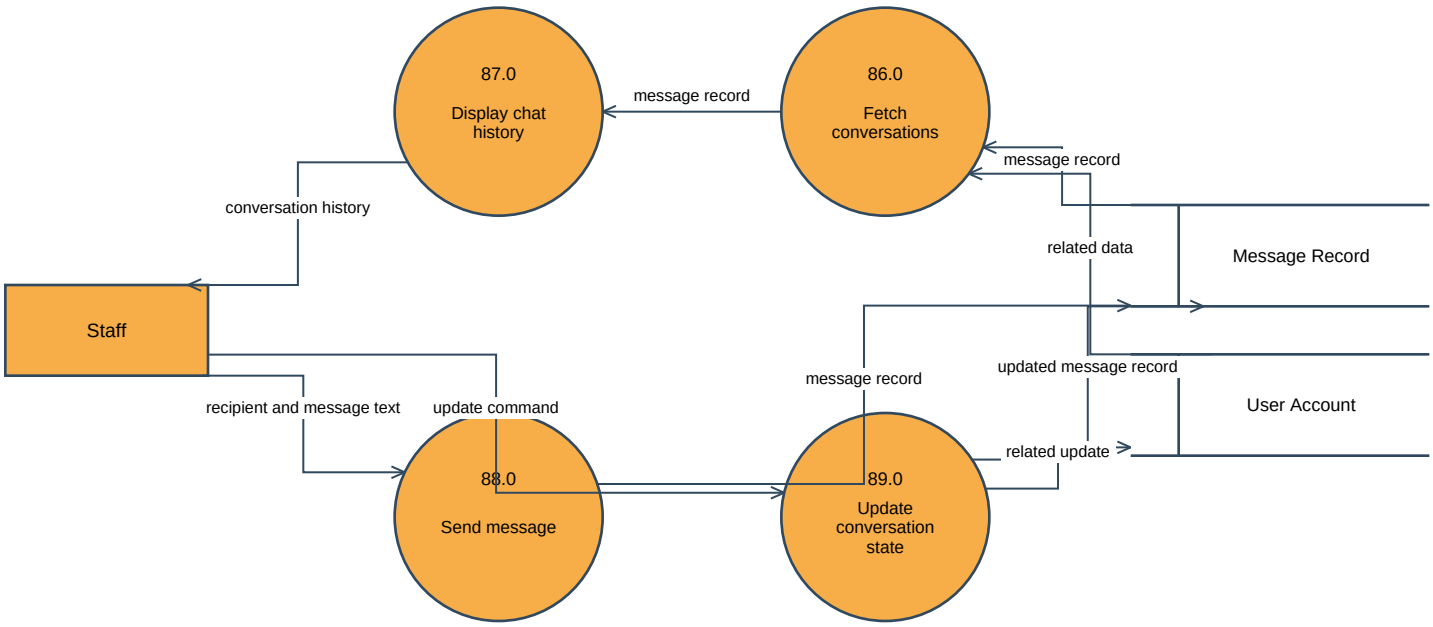
Staff Payment Management and Reporting

Staff record and review payments and generate operational reports from appointment, service, payment, and inventory information. The system returns report results and stores the latest payment status for clinic accounting.



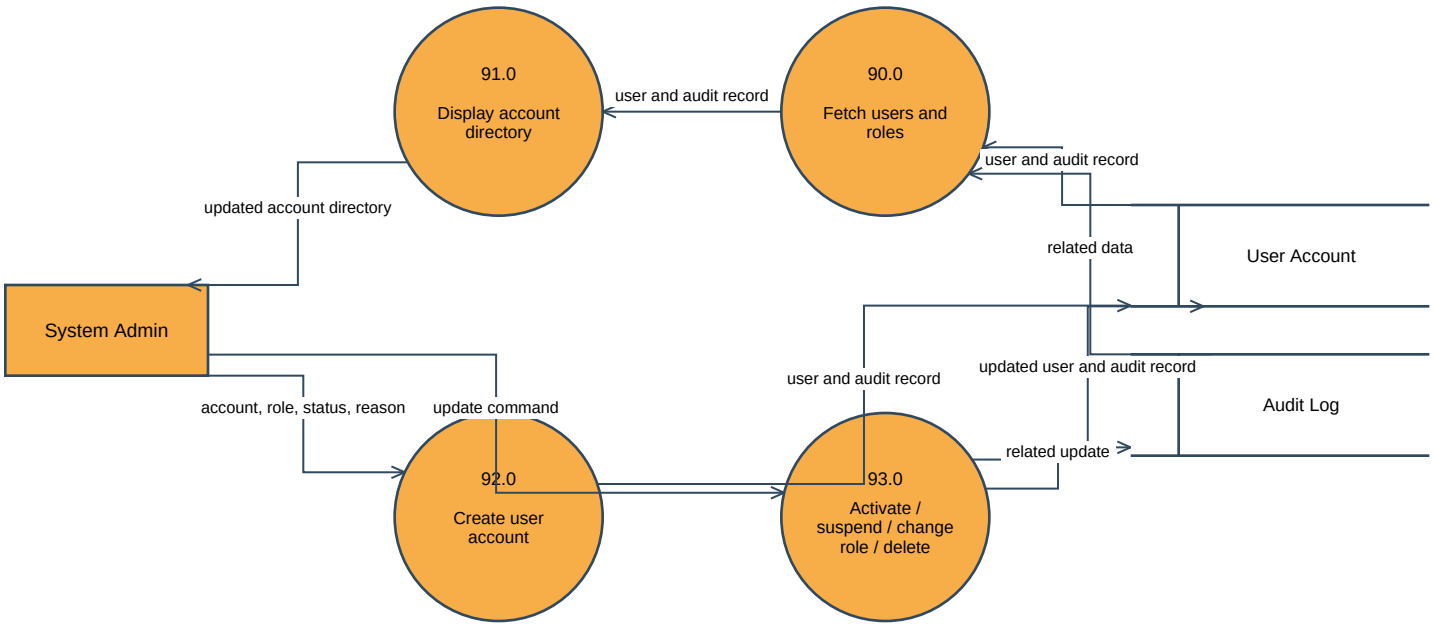
Staff Messaging

Staff retrieve user conversations, open a selected thread, and send messages. Each message is stored with sender, recipient, conversation, and time information so the complete communication history can be displayed and traced.



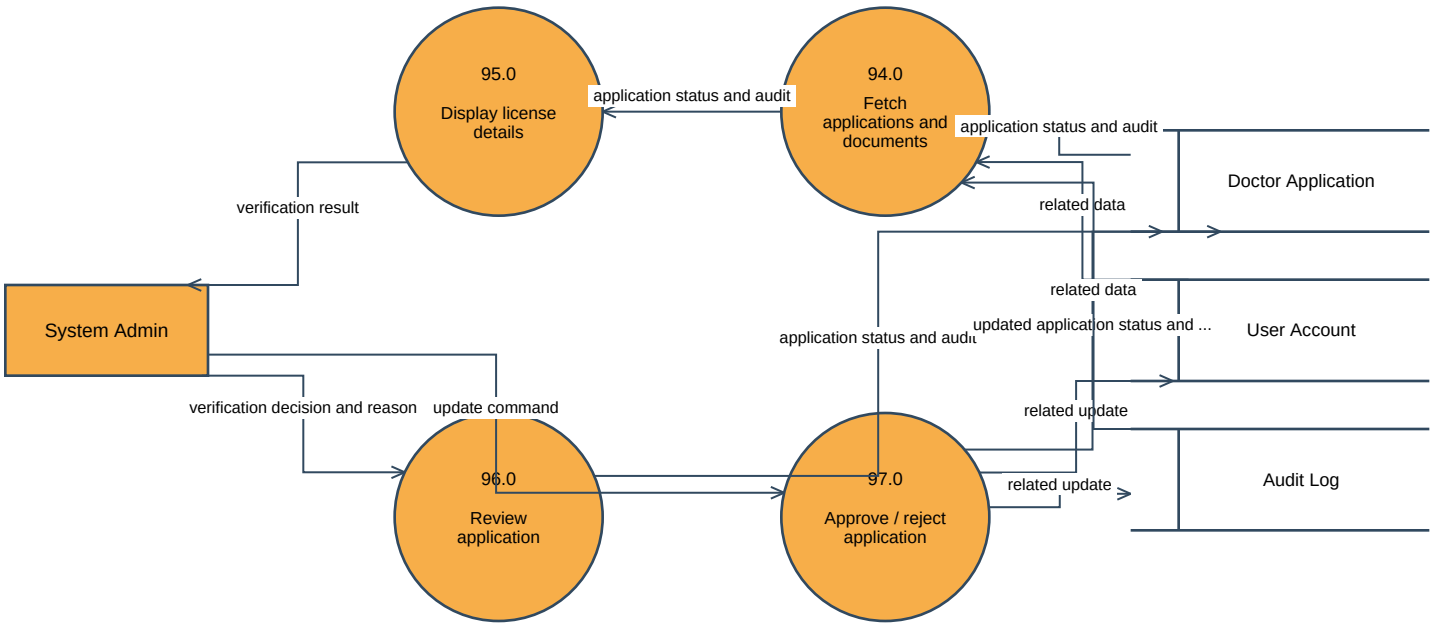
System Administrator User and Role Management

The System Administrator creates accounts, searches and views users, assigns roles, activates or suspends accounts, changes account roles, and deletes accounts. Every sensitive action is recorded in the audit log.



System Administrator Doctor Verification

The System Administrator retrieves pending doctor applications, reviews contact details, qualifications, license number, expiry date, specialty, and submitted documents, and then approves or rejects the application with a recorded reason.



System Dashboard, Alerts, Reports, and Audit Logs

The System Administrator monitors totals for users, doctors, staff, owners, appointments, and queue entries; reviews doctor-verification, emergency, low-stock, and expiry alerts; and searches the append-only audit trail of sensitive administrative actions.

